

Max-Flow Min-Cut Theorem

Abstract

We develop the theory of flows in networks from first principles and give a detailed proof of the Max-Flow Min-Cut Theorem. We introduce flow networks, cuts, residual networks, and augmenting paths, and we prove all intermediate lemmas needed to establish the equivalence between maximal flows and minimum s - t cuts. We also derive some fundamental corollaries, including the integrality of maximum flows with integral capacities.

1 Flow Networks

We begin by formalizing the central objects of study.

Definition 1.1 (Directed graph). A *directed graph* is a pair $G = (V, E)$ where V is a finite set of *vertices* and $E \subseteq V \times V$ is a set of ordered pairs called *directed edges* or simply *edges*. If $(u, v) \in E$ we say there is an edge from u to v and write $u \rightarrow v$.

Definition 1.2 (Flow network). A *flow network* is a quadruple

$$N = (G, c, s, t)$$

where

- $G = (V, E)$ is a finite directed graph,
- $c : E \rightarrow \mathbb{R}_{\geq 0}$ is a *capacity function* assigning to each edge $e \in E$ a nonnegative real number $c(e)$,
- $s \in V$ is a distinguished vertex called the *source*, and
- $t \in V$ is a distinguished vertex called the *sink*, with $s \neq t$.

We assume there are no edges entering s and no edges leaving t ; this can always be enforced by standard transformations without changing the theory.

We interpret $c(u, v)$ as the maximum amount of “flow” that can be sent along edge (u, v) per unit time.

Definition 1.3 (Flow). Let $N = (G, c, s, t)$ be a flow network with $G = (V, E)$. A *flow* is a function $f : E \rightarrow \mathbb{R}_{\geq 0}$ satisfying

(i) **Capacity constraints.** For every edge $e \in E$,

$$0 \leq f(e) \leq c(e).$$

(ii) **Flow conservation.** For every vertex $v \in V \setminus \{s, t\}$,

$$\sum_{(u,v) \in E} f(u, v) = \sum_{(v,w) \in E} f(v, w).$$

That is, total inflow equals total outflow at every intermediate vertex.

Definition 1.4 (Value of a flow). Let f be a flow. The *value* of f is the total flow leaving the source, defined by

$$\text{val}(f) := \sum_{(s,v) \in E} f(s, v).$$

When the network has no edges entering s and no edges leaving t , this is also equal to the total flow entering t ; we prove this formally below.

1.1 Cuts and their capacities

Definition 1.5 (s - t cut). Let $N = (G, c, s, t)$ be a flow network. An s - t cut is an ordered pair (S, T) of subsets of V such that

$$S \cup T = V, \quad S \cap T = \emptyset, \quad s \in S, \quad t \in T.$$

We often implicitly write $T = V \setminus S$ and speak of the cut “ S ”.

Intuitively, an s - t cut is a partition of the vertices separating s from t . We measure how “restrictive” a cut is by the total capacity of the edges that cross it.

Definition 1.6 (Capacity of a cut). Let (S, T) be an s - t cut. The *capacity* of (S, T) is

$$c(S, T) := \sum_{\substack{(u,v) \in E \\ u \in S, v \in T}} c(u, v).$$

Only edges directed from S to T contribute.

Definition 1.7 (Minimum s - t cut). The *minimum s - t cut value* is

$$\lambda^* := \min\{c(S, T) : (S, T) \text{ is an } s\text{-}t \text{ cut}\}.$$

Any cut (S, T) with $c(S, T) = \lambda^*$ is called a *minimum cut*.

2 The Flow Value Lemma and Cut Upper Bound

We first show that the value of any flow is bounded above by the capacity of any s - t cut. The key technical ingredient is the following lemma.

Lemma 2.1 (Flow value lemma). *Let f be a flow in $N = (G, c, s, t)$. For any subset $X \subseteq V$ with $s \in X$ and $t \notin X$, we have*

$$\text{val}(f) = \sum_{\substack{(u,v) \in E \\ u \in X, v \notin X}} f(u, v) - \sum_{\substack{(u,v) \in E \\ u \notin X, v \in X}} f(u, v).$$

In words: the value of the flow equals the net flow crossing any “ s -side” vertex set X .

Proof. Define the *net outflow* of f from X by

$$F(X) := \sum_{\substack{(u,v) \in E \\ u \in X, v \notin X}} f(u, v) - \sum_{\substack{(u,v) \in E \\ u \notin X, v \in X}} f(u, v).$$

We claim $F(X) = \text{val}(f)$ for any X with $s \in X$ and $t \notin X$.

Rewrite $F(X)$ by grouping by the head or tail vertex:

$$\begin{aligned} F(X) &= \sum_{u \in X} \sum_{\substack{(u,v) \in E \\ v \notin X}} f(u, v) - \sum_{v \in X} \sum_{\substack{(u,v) \in E \\ u \notin X}} f(u, v) \\ &= \sum_{v \in X} \left(\sum_{\substack{(v,w) \in E \\ w \notin X}} f(v, w) - \sum_{\substack{(u,v) \in E \\ u \notin X}} f(u, v) \right), \end{aligned}$$

where in the second line we just renamed the dummy variables.

Now split the sum over $v \in X$ into $v = s$ and $v \in X \setminus \{s\}$:

$$F(X) = \left(\sum_{\substack{(s,w) \in E \\ w \notin X}} f(s, w) - \sum_{\substack{(u,s) \in E \\ u \notin X}} f(u, s) \right) + \sum_{v \in X \setminus \{s\}} \left(\sum_{\substack{(v,w) \in E \\ w \notin X}} f(v, w) - \sum_{\substack{(u,v) \in E \\ u \notin X}} f(u, v) \right).$$

By assumption, there are no edges entering s , so the sum over (u, s) is zero, and the first term reduces to

$$\sum_{(s,w) \in E} f(s, w) = \text{val}(f),$$

since $w \notin X$ automatically (because $s \in X$ and (s, w) can only leave X).

Next consider any $v \in X \setminus \{s\}$. Split all edges into those with head v or tail v and those with both endpoints in X or both in $V \setminus X$. A short check (or a picture) shows that

$$\sum_{\substack{(v,w) \in E \\ w \notin X}} f(v, w) - \sum_{\substack{(u,v) \in E \\ u \notin X}} f(u, v) = \sum_{(v,w) \in E} f(v, w) - \sum_{(u,v) \in E} f(u, v),$$

because internal edges with both endpoints in X cancel between the two sums, and edges with both endpoints outside X do not appear at all.

By flow conservation at each $v \in V \setminus \{s, t\}$, we have

$$\sum_{(v,w) \in E} f(v, w) = \sum_{(u,v) \in E} f(u, v),$$

so the expression for each $v \in X \setminus \{s\}$ is 0. Therefore

$$F(X) = \text{val}(f) + \sum_{v \in X \setminus \{s\}} 0 = \text{val}(f),$$

which is the desired equality. $\square$

The upper bound on flow value by cut capacity is now immediate.

Proposition 2.2 (Cut upper bound). *Let f be a flow and (S, T) an s - t cut. Then*

$$\text{val}(f) \leq c(S, T).$$

Proof. Apply Lemma 2.1 with $X = S$. We obtain

$$\text{val}(f) = \sum_{(u,v) \in S \times T} f(u, v) - \sum_{(u,v) \in T \times S} f(u, v).$$

The second sum is nonnegative, since $f(u, v) \geq 0$ for all edges. Therefore

$$\text{val}(f) \leq \sum_{(u,v) \in S \times T} f(u, v) \leq \sum_{(u,v) \in S \times T} c(u, v) = c(S, T),$$

where the second inequality uses the capacity constraints $f(u, v) \leq c(u, v)$. $\square$

As an immediate corollary, the value of any flow is bounded above by the value of any cut, and thus also by the minimum cut value.

Corollary 2.3. *For every flow f ,*

$$\text{val}(f) \leq \lambda^*.$$

Proof. For any s - t cut (S, T) , Proposition 2.2 gives $\text{val}(f) \leq c(S, T)$. Taking the minimum over all cuts yields

$$\text{val}(f) \leq \min\{c(S, T)\} = \lambda^*.$$

$\square$

The Max-Flow Min-Cut Theorem will assert that this upper bound is tight: there exists a flow whose value is *equal* to λ^* .

3 Residual Networks and Augmenting Paths

To show that the upper bound is tight, we need a way to *improve* a given flow if it is not optimal. This is achieved via residual networks and augmenting paths.

Definition 3.1 (Residual capacity). Let f be a flow. For an edge $(u, v) \in E$, its *residual capacity* with respect to f is

$$c_f(u, v) := c(u, v) - f(u, v).$$

In addition, for every edge $(u, v) \in E$, we introduce a *reverse edge* (v, u) whose residual capacity is

$$c_f(v, u) := f(u, v).$$

If $(u, v) \notin E$ we set $c(u, v) = 0$ and we may treat $c_f(u, v)$ as defined accordingly (possibly zero).

The intuition is that $c_f(u, v)$ measures how much additional flow we can send forward on (u, v) , while $c_f(v, u)$ measures how much flow we can “cancel” on (u, v) by sending flow back along (v, u) .

Definition 3.2 (Residual network). Given a flow f , the *residual network* $G_f = (V, E_f)$ is the directed graph with vertex set V and edge set

$$E_f := \{(u, v) \in V \times V : c_f(u, v) > 0\},$$

where c_f is the residual capacity function defined above.

Definition 3.3 (Augmenting path). An *augmenting path* for f is a directed path P from s to t in the residual network G_f . The *residual capacity* of P is

$$\Delta_f(P) := \min\{c_f(u, v) : (u, v) \text{ is an edge of } P\}.$$

3.1 Augmenting along a path

We next formalize the process of increasing a flow along an augmenting path.

Lemma 3.1 (Augmenting a flow on a path). Let f be a flow and let P be an s – t augmenting path in G_f with residual capacity $\Delta = \Delta_f(P) > 0$. Define $f' : E \rightarrow \mathbb{R}_{\geq 0}$ by

$$f'(u, v) := \begin{cases} f(u, v) + \Delta, & \text{if } (u, v) \text{ is a forward edge of } P, \\ f(u, v) - \Delta, & \text{if } (v, u) \text{ is a forward edge of } P, \\ f(u, v), & \text{otherwise.} \end{cases}$$

Then f' is also a flow, and

$$\text{val}(f') = \text{val}(f) + \Delta.$$

Proof. First we show that f' satisfies the capacity constraints. Consider any edge $(u, v) \in E$.

- If (u, v) is not used (in either direction) by P , then $f'(u, v) = f(u, v)$, and the capacity constraints are inherited from f .
- If (u, v) is a forward edge on P , then $(u, v) \in E_f$ and $c_f(u, v) = c(u, v) - f(u, v) \geq \Delta$. Hence

$$f'(u, v) = f(u, v) + \Delta \leq f(u, v) + c_f(u, v) = c(u, v).$$

Moreover $f'(u, v) \geq 0$ since $f(u, v) \geq 0$ and $\Delta \geq 0$.

- If (v, u) is a forward edge on P , then by definition of E_f we have

$$c_f(v, u) = f(u, v) \geq \Delta.$$

Thus $f(u, v) \geq \Delta$ and

$$f'(u, v) = f(u, v) - \Delta \geq 0.$$

Also $f'(u, v) \leq f(u, v) \leq c(u, v)$, so the upper capacity bound holds.

Thus $0 \leq f'(u, v) \leq c(u, v)$ for all edges.

Next we check flow conservation at each vertex $x \in V \setminus \{s, t\}$. Only edges in P incident to x can change their flow values; all other incident edges keep the same flow. On the path P , the vertex x has either degree 0 (if it is not on P), degree 2 as an internal vertex, or degree 1 as s or t . For $x \notin \{s, t\}$ there are only two possibilities:

- If x is not on P , then no incident flows change, so conservation holds since it held for f .
- If x is an internal vertex of P , then exactly two incident edges of x appear in P : one entering and one leaving. Whether each is a forward or backward use of the underlying edge in G , the change in inflow at x is exactly $\pm\Delta$ and the change in outflow is the same $\pm\Delta$, so the net change is 0. Thus conservation at x is preserved.

Therefore f' satisfies flow conservation at every intermediate vertex.

Finally, we examine the value. Consider the net flow out of s . Only edges in P incident to s change their flow, and exactly one such edge is used on P . If the first edge of P is (s, v) , then it is used in the forward direction and we increase $f(s, v)$ by Δ , increasing $\text{val}(f)$ by Δ . If the first edge is (v, s) used in the forward direction, then we decrease $f(v, s)$ by Δ , which by conservation implies that net flow out of s increases by Δ as well (intuitively, we are sending Δ less flow into s , which corresponds to Δ more units net leaving s). A more formal argument can be made using Lemma 2.1 applied to $X = \{s\}$, for which $F(X)$ counts exactly the net outflow from s .

In either case, we conclude

$$\text{val}(f') = \text{val}(f) + \Delta,$$

as claimed. □

4 Characterization of Optimality

We now prove a fundamental characterization: a flow is maximal if and only if its residual network has no s - t path. This is the core of the Max-Flow Min-Cut Theorem.

Lemma 4.1 (No augmenting path $\Rightarrow$ cut of same value). *Let f be a flow and suppose the residual network G_f contains no directed path from s to t . Let $S \subseteq V$ be the set of vertices reachable from s in G_f , and let $T := V \setminus S$. Then:*

- (a) (S, T) is an s - t cut.
- (b) For every edge (u, v) with $u \in S$ and $v \in T$, we have $f(u, v) = c(u, v)$ (such edges are saturated).
- (c) For every edge (u, v) with $u \in T$ and $v \in S$, we have $f(u, v) = 0$ (such edges carry no flow).
- (d) $\text{val}(f) = c(S, T)$.

Proof. (a) By definition, S is the set of vertices reachable from s in G_f . Thus $s \in S$. Since there is no path from s to t in G_f , the sink t is not reachable, so $t \in T$, and (S, T) is indeed an s - t cut.

- (b) Suppose $u \in S$, $v \in T$, and $(u, v) \in E$. If $c_f(u, v) > 0$ then $(u, v) \in E_f$ and hence v would be reachable from s in G_f (since u is reachable and there is an edge from u to v), which would force $v \in S$, contradicting $v \in T$. Therefore

$$c_f(u, v) = c(u, v) - f(u, v) = 0 \quad \Rightarrow \quad f(u, v) = c(u, v).$$

- (c) Suppose $u \in T$, $v \in S$, and $(u, v) \in E$. If $f(u, v) > 0$, then the reverse edge (v, u) in the residual network has

$$c_f(v, u) = f(u, v) > 0,$$

so $(v, u) \in E_f$ and hence u would be reachable from s in G_f (because $v \in S$), implying $u \in S$. This contradicts $u \in T$. Therefore $f(u, v) = 0$.

- (d) Applying Lemma 2.1 with $X = S$ we obtain

$$\text{val}(f) = \sum_{\substack{(u,v) \in E \\ u \in S, v \in T}} f(u, v) - \sum_{\substack{(u,v) \in E \\ u \in T, v \in S}} f(u, v).$$

By (c), every edge from T to S carries zero flow, so the second sum is 0. By (b), every edge from S to T is saturated, so $f(u, v) = c(u, v)$ for such edges. Therefore

$$\text{val}(f) = \sum_{(u,v) \in S \times T} c(u, v) = c(S, T).$$

□

Lemma 4.2 (Characterization via augmenting paths). *Let f be a flow in N . Then the following are equivalent:*

- (i) f is a maximum flow (i.e., $\text{val}(f)$ is as large as possible).
- (ii) There is no s - t augmenting path in G_f .
- (iii) There exists an s - t cut (S, T) such that $\text{val}(f) = c(S, T)$.

Proof. We show (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii): Suppose f is a maximum flow but G_f contains an s - t augmenting path P . By Lemma 3.1, we can construct a flow f' with $\text{val}(f') = \text{val}(f) + \Delta$ for some $\Delta > 0$, contradicting maximality of f . Hence no augmenting path exists.

(ii) $\Rightarrow$ (iii): Suppose there is no augmenting path. By Lemma 4.1, the sets S and T defined as the vertices reachable and unreachable (respectively) from s in G_f form an s - t cut with $\text{val}(f) = c(S, T)$.

(iii) $\Rightarrow$ (i): Suppose there exists an s - t cut (S, T) with $\text{val}(f) = c(S, T)$. For any flow g , by Proposition 2.2 we have

$$\text{val}(g) \leq c(S, T) = \text{val}(f),$$

so $\text{val}(f)$ is an upper bound on the value of any flow. Thus f is a maximum flow. $\square$

5 The Max-Flow Min-Cut Theorem

We are now ready to state and prove the main result.

Theorem 5.1 (Max-Flow Min-Cut Theorem). *In any flow network $N = (G, c, s, t)$, the maximum value of an s - t flow equals the minimum capacity of an s - t cut. More precisely,*

$$\max\{\text{val}(f) : f \text{ is a flow in } N\} = \min\{c(S, T) : (S, T) \text{ is an } s\text{-}t \text{ cut}\}.$$

Proof. Let f^* be any maximum flow (which exists since the set of feasible flows is nonempty and bounded, and the value is a continuous function on a compact polytope). By Corollary 2.3 we have

$$\text{val}(f^*) \leq \lambda^*,$$

where λ^* is the minimum cut capacity.

By Lemma 4.2, there exists an s - t cut (S^*, T^*) such that

$$\text{val}(f^*) = c(S^*, T^*).$$

Since $c(S^*, T^*) \geq \lambda^*$ by definition of λ^* , we deduce

$$\text{val}(f^*) = c(S^*, T^*) \geq \lambda^*.$$

Combining the two inequalities yields

$$\text{val}(f^*) = \lambda^*,$$

which is exactly the claimed equality. $\square$

The standard picture is now complete: any maximum flow and any minimum cut have the same value; furthermore, the structure of the residual network at optimality witnesses a particular minimum cut.

6 Algorithmic Corollaries

The proof of the Max-Flow Min-Cut Theorem via augmenting paths directly motivates the *Ford–Fulkerson method*, an algorithmic procedure for finding a maximum flow.

Definition 6.1 (Ford–Fulkerson method (informal)). Starting from the zero flow $f \equiv 0$, repeatedly find an s – t augmenting path P in the residual network G_f , augment the flow by a positive amount Δ along P as in Lemma 3.1, and update f and G_f . The method terminates when no augmenting path exists.

By Lemma 4.2, any flow at which no augmenting path exists is maximum. This yields:

Corollary 6.1 (Correctness of Ford–Fulkerson). *If the Ford–Fulkerson method terminates at a flow f , then f is a maximum flow, and its value equals the capacity of some minimum s – t cut.*

Proof. Upon termination there is no s – t path in the residual network G_f . By Lemma 4.2, f is a maximum flow and there exists a cut (S, T) with $\text{val}(f) = c(S, T) = \lambda^*$. $\square$

6.1 Integrality of maximum flows

A particularly important consequence is integrality: if capacities are integers, then there is an optimal flow which is also integral.

Theorem 6.2 (Integrality theorem). *Let $N = (G, c, s, t)$ be a flow network where all capacities $c(e)$ are integers. Assume that the Ford–Fulkerson method always chooses an s – t augmenting path P and augments by $\Delta = \Delta_f(P)$ as in Lemma 3.1. Then:*

- (a) *All flows encountered during the algorithm assign integer values to every edge.*
- (b) *The algorithm terminates after finitely many augmentations.*
- (c) *The final flow is a maximum flow with integer values on each edge.*

Proof. (a) We prove by induction on the number of augmentations. Initially, the zero flow is integral. Suppose the current flow f is integral. Then every residual capacity $c_f(u, v)$ is an integer, being the difference of integers. In particular, the residual capacity Δ of any augmenting path P is the minimum of finitely many integers and hence an integer. By Lemma 3.1, each updated value $f'(u, v)$ differs from $f(u, v)$ by either 0 or $\pm\Delta$, so $f'(u, v)$ is still an integer. Thus integrality is preserved.

(b) At each augmentation, the value of the flow $\text{val}(f)$ increases by $\Delta \geq 1$, an integer. On the other hand, by Corollary 2.3, $\text{val}(f) \leq \lambda^*$ for every flow f . Because capacities are integers, λ^* is also an integer. Hence the value can increase at most λ^* times before reaching its upper bound. Therefore, there can be at most λ^* augmentations, and the algorithm terminates after finitely many steps.

(c) By Corollary 6.1, the terminating flow is maximum. By part (a), it is integral.



7 Further Consequences and Remarks

We briefly mention a classical combinatorial corollary, obtained by taking unit capacities; we state it without proof, as its derivation is now standard.

Corollary 7.1 (Edge-disjoint paths vs. edge cuts). *Let $G = (V, E)$ be a directed graph with distinct vertices $s, t \in V$. Then the maximum number of pairwise edge-disjoint directed s - t paths equals the minimum size of a set of edges whose removal disconnects t from s .*

Remark 7.1. This is a special case of a more general set of min-max theorems in combinatorial optimization, such as Menger's theorem for vertex-disjoint paths and vertex cuts, and the max-flow min-cut theorem is one of the foundational examples of such dualities.